

Rendy Ramadhana Wahyudi

INSTRUMENTATION & FIELD ENGINEERING | ENERGY SERVICE READINESS | OILFIELD ENTHUSIAST

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PROFESSIONAL SUMMARY

Instrumentation-focused graduate with strong field experience in geophysics and advanced sensor systems. Dedicated to integrating hardware-software solutions for efficient and reliable field operations in energy services. Comfortable working in remote conditions, committed to safety, operational excellence, and dependable team coordination in dynamic field environments.

CORE COMPETENCIES

✓ Instrumentation	✓ Data Acquisition	✓ System Maintenance
✓ Sensors	✓ Control System	✓ Signal Processing
✓ Electronics	✓ Scientific Computing	✓ Embedded Programming

PROFESSIONAL EXPERIENCE

Geophysics Technician (Contract-Based), PT. Surya Brinka Persada, Bandung | Sep 2024 – Present

- Led 2D resistivity geophysical surveys to support groundwater potential assessments across West Java.
- Coordinated a three-member field team to conduct geophysical surveys at 16 dispersed locations, optimizing technical role assignments to significantly reduce survey duration and operational costs.
 - Completed all projects ahead of schedule, delivering acquisition data with less than 10% error—accepted by stakeholders without the need for revision.

Instrumentation Engineer (Contract-Based), Geophysics Research Team, UNPAD, Sumedang | Jul – Aug 2023

- Developed systems under tight deadlines, applying various approaches to enhance operational effectiveness.
- Initiated Instrumentation performance optimization using open-source numerical simulation software, resulting in up to 50% savings in cost and material use.
 - Designed and built a functional prototype that served as the foundation for measurable and sustainable further research development.

LEADERSHIP & FIELD COORDINATION EXPERIENCE

Village Empowerment Program Coordinator, HIMA Geofisika PEDRA, UNPAD | Sep 2021 – Feb 2022

- Coordinated a 10-member team for a long-term community development program reaching over 200 residents.
- Organized earthquake preparedness training using the CBDRR approach and conducted emergency response mock drills to improve information dissemination effectiveness.
 - Successfully increased community disaster awareness by 80% based on post-activity surveys.

FIELD READINESS & SAFETY

- Eager to learn and gain hands-on experience in remote, challenging environments.
- Experienced working independently under minimal supervision, with a strong focus on safety standards.
- Physically and mentally prepared for long hours and demanding field conditions.

EDUCATION

Master of Science in Physics, Universitas Padjadjaran, West Java | Jan 2023 – Aug 2025 (Expected)

- Capstone Project:** *Development of a disaster monitoring system integrating environmental sensors, real-time data acquisition, and embedded control for regional early-warning applications.*

Bachelor of Science in Geophysical, Universitas Padjadjaran, West Java | Aug 2019 – Jul 2023

- Capstone Project:** *Prototype development of IoT-based distributed seismic sensor network*, published at DOI: 10.2991/978-2-38476-283-5_7.

CERTIFICATES & GRANTS

Graduate Scholarship Awardee, JFLS, Disdik Jabar | Feb 2025

Graduate Research Grant, Kemdiktisaintek | Aug 2024

TOEFL ITP Score: 527, Conversational English, LBI FIB Universitas Indonesia | Jul 2024

TECHNICAL SKILLS

Instrumentation: Sensor Integration, STM32, ESP32 | **Embedded Systems:** C++, Arduino, STM32CubeIDE, KiCad | **Signal & Data Processing:** Python (NumPy, SciPy), MATLAB, Machine learning, Big Data | **Simulation & Testing Tools:** Proteus, LabVIEW | **Documentation:** LaTeX, MS Word | **Field Tools & OS:** Linux (basic terminal ops), Windows, Microosft Teams, Primavera P6.